MOTILAL NEHRU NATIONAL INSTITUTE OF TECHNOLOGY, ALLAHABAD

DATA STRUCTURE LAB

ASSIGNMENT- 4 (Linked List: Singly)

1. Write WAP to implement Linked List.
    a. Create a linked list.
    b. Insert an element at the start of the linked list.
    c. Insert an element at the end of the linked list.
    d. Insert an element before an existing element whose information is *x* in a linked list.
    e. Insert an element after an existing element whose information is *x* in a linked list.
    f. Delete the first element of the linked list.
    g. Delete the last element of the linked list.
    h. Delete the element whose information is *x* from a linked list.
    i. Display the contents of the linked list.
2. In a college library, you are tasked with implementing a feature to manage a list of books represented as a linked list, where each node contains details of a book (like title, author) and a reference to the next book. You need to write a recursive function, that takes the head of the linked list as input and returns the total number of books (i.e., the length of the linked list). The function should use recursion to traverse the list and count the number of nodes, returning 0 when the end of the list (null or none) is reached.
3. In a bank there are customers, you have a sorted linked list storing customer IDs. Sometimes, due to data entry errors, duplicate customer IDs are added to the list. You need to implement a function to remove any duplicate customer IDs from the sorted linked list, ensuring each ID appears only once.
4. In an inventory management system, you have two separate linked lists representing different categories of products. You need to merge these two linked lists into one, maintaining the order of products in both lists. Write a function concatenate_lists(list1, list2) to concatenate the two linked lists. Ensure the products from both categories are combined into a single linked list.
5. In a task scheduling system, you maintain a linked list of tasks, where each node represents a task and its priority. To optimize the schedule, you need to remove every alternate task from the list, effectively deleting tasks that are not critical. Write a function delete_alternate_nodes(head) to delete every alternate node in the linked list. The function should update the list by removing every second task. Ensure the solution works for any size of the linked list.
6. In a document management system, you store the titles of books in a linked list, with each node containing a book title. You need to check if the list of book titles forms a palindrome, meaning the titles are the same when read forwards and backwards. Write a function to determine if the linked list of book titles is a palindrome. The function should return true if the titles are the same in both directions, and False otherwise. Ensure that the solution works with any length of linked list.
7. In a task management system, you store a list of tasks in a linked list, with each node representing a task and its priority. You want to reverse the order of tasks, so that the last task added appears first and the first task appears last. Write a function that should rearrange the nodes in reverse order without creating a new list. Make sure the solution works efficiently for any number of tasks.